CODE SPORT

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In this month's column, we discuss a few computer science interview questions.

This month, let's discuss a set of computer science interview questions, focusing on natural language processing, machine learning as well as on traditional programming problems. Please note that it is difficult to discuss the complete solution to the questions due to the large number of questions we cover in this column. Instead, I encourage you to send me your solutions to the questions posed, directly, so that I can provide feedback on the same.

1. You are given the problem of disambiguating entities in a given set of documents. For example, for the word 'Apple', you will need to disambiguate between the use of the word 'Apple' as an corporate entity and its use as a word that describes a fruit. A more complicated example would be the following:

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S1: The White House announced its intention to support the newly formed government in Vanaru.
S2: The New Year Party will be held at the White House, and will be hosted by Michelle Obama.
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In sentence S1, the White House is an entity that actually represents the American government. In sentence S2, the White House is a location. Can you come up with an algorithm which can disambiguate such mentions of this entity to their appropriate types, based on the context.

2. In information retrieval, you are given a set of documents and a query. You are asked to retrieve the set of documents that is most relevant to the query. Many of the common information retrieval techniques are built around the measure of TF-IDF, where TF stands for the term frequency, i.e., the number of times a query keyword appears in a document. For example, given the query 'Tendulkar the Indian Cricketer', I am asked to find all documents relevant to this query. It is obvious that one can compute the term frequency for the query terms, and return a ranked list of documents in which the query terms appear most frequently. If this suffices to return the most relevant documents, why do we need the measure of the inverse document frequency (IDF)? Can you illustrate with an example why IDF is needed in information retrieval?
3. Parsing is one of the major components of a natural language processing pipeline. Depending on the nature of the task, different types of parsing of the input text may be required. Can you explain the differentiation between shallow parsing, dependency parsing and constituency parsing? Given the sentence, "John went to New York yesterday by the midnight flight," can you provide the output for each of these parses? On a related note, when would you use POS tagging instead of any parsing?
4. You have been asked to build an application that analyses the conversations in call centres. These conversations typically happen between the agent and the customer. An automatic speech recognition system has been used to convert the voice conversations into text. The first stage of your application needs to analyse the voice transcripts, and correct any mistakes in the words that would have been caused by the automatic speech recognition system. Can you explain how you would build the transcript correction system?
5. An easier and related problem to the question